Inverted MAML Card 1: Round 4 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.045, L2=2048, Mom=0.88, Clamp=±75 LSB

1. Round 4 Summary & Cumulative Comparison (R1–R4)

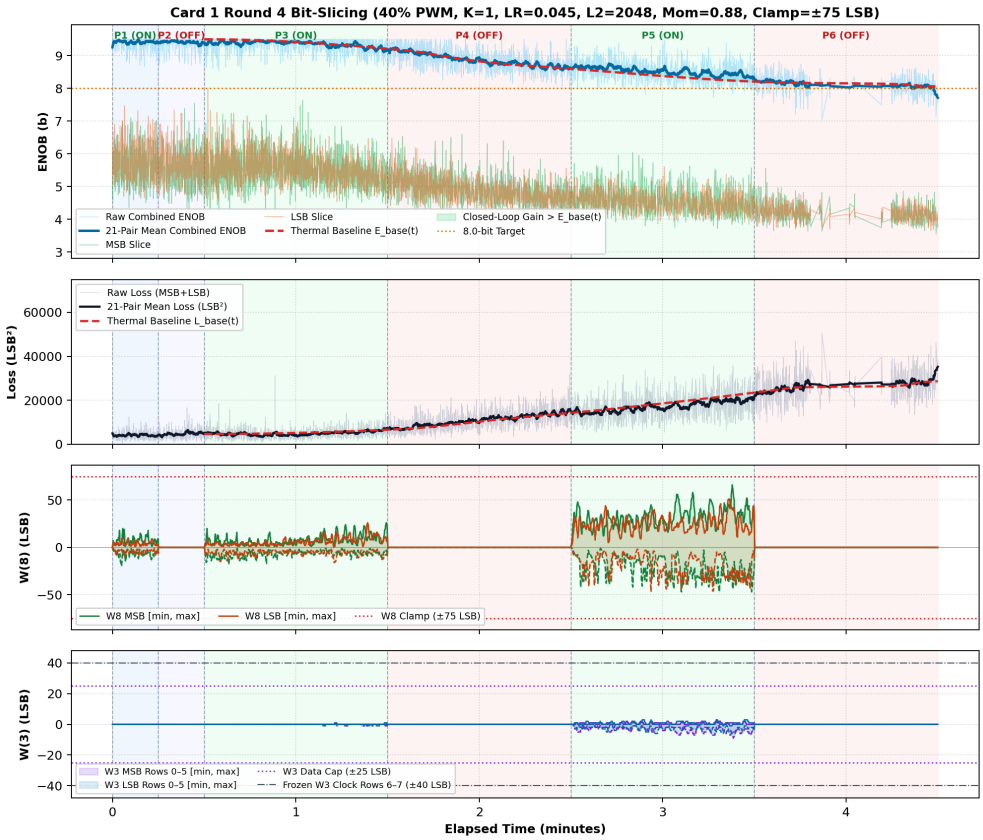
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 4 Detrended Trend Metrics:** Lift vs E_base = **+0.11 b**, Loss Ratio = **1.17x**, Climb = **+0.14 b**, Jitter = **0.24 b**, Recovery = **1.00**, Score = **+0.643**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R1	0.03	1024	0.9	±75	+0.66b	1.57x	+0.74b	0.33b	0.67	+1.957
R2*	0.035	1536	0.85	±75	+0.50b	1.38x	+0.82b	0.29b	0.96	+2.004
R3	0.02	2560	0.8	±70	+0.09b	1.11x	+0.22b	0.26b	1.00	+0.559
R4	0.045	2048	0.88	±75	+0.11b	1.17x	+0.14b	0.24b	1.00	+0.643

2. Phase-by-Phase Telemetry (3,837 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	328	9.44->9.42 (9.42b)	-	3711->3954	[20,11]	0	328
P2	Cold / Regler OFF	303	9.39->9.39 (9.41b)	-	4144->5037	[0,0]	0	0
P3	Warm / Regler ON	1055	9.35->9.28 (9.36b)	-0.02 b	5087->5570	[26,26]	1	1055
P4	Warm / Regler OFF	902	9.14->8.70 (8.88b)	+0.03 b	7340->13333	[0,0]	0	0
P5	Warm / Regler ON	793	8.65->8.42 (8.53b)	+0.16 b	14093->19402	[66,51]	9	793
P6	Warm / Regler OFF	456	8.21->8.05 (8.12b)	-0.04 b	23749->28367	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 4

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±75 LSB, while W(3) rows 0–5 stay within ±25 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 4 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

